

Assignment 9

1. Create an index that will enable a user to pull orders for '1990-10-03' out of the Orders table quickly.

```

D6_99024_tejas@show index from orders;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name      | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null |
| Index_type | Comment | Index_comment | Visible | Expression |          |            |          |        |      |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| orders |          1 | order_date_index |          1 | odate       |          |          2 |          |        |      |
| BTREE   |          |                  | YES        | NULL        |          |            |          |        |      |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
D6_99024-tejas>alter table orders add unique (onum);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

3. Create an index that would permit salesperson to retrieve his orders.

```
create index index_salesperson on orders(snum);
```

```
D6_99024_tejas>show index from orders;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null
Index_type	Comment	Index_comment	Visible	Expression					
orders	0	onum	1	onum	A	10	NULL	NULL	YES
BTREE			YES	NULL					
orders	1	order_date_index	1	odate	A	2	NULL	NULL	YES
BTREE			YES	NULL					
orders	1	index_salesperson	1	snum	A	5	NULL	NULL	YES
BTREE			YES	NULL					

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
D6_99024_tejas>create unique index index_salesperson on customers(snum,cname,rating);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

D6_99024_tejas>show index from customers
-> ;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null
Index_type	Comment	Index_comment	Visible	Expression					
customers	0	index_salesperson	1	snum	A	5	NULL	NULL	YES
S BTREE			YES	NULL					
customers	0	index_salesperson	2	cname	A	7	NULL	NULL	YES
S BTREE			YES	NULL					
customers	0	index_salesperson	3	rating	A	7	NULL	NULL	YES
S BTREE			YES	NULL					

5. Create an index to speed up searching order on a given date by given customer.

create index index_order on orders(cnum,odate);

```
D6_99024_tejas>show index from orders;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null
Index_type	Comment	Index_comment	Visible	Expression					
orders	0	onum	1	onum	A	10	NULL	NULL	YES
BTREE			YES	NULL					
orders	1	order_date_index	1	odate	A	2	NULL	NULL	YES
BTREE			YES	NULL					
orders	1	index_salesperson	1	snum	A	5	NULL	NULL	YES
BTREE			YES	NULL					
orders	1	index_order	1	cnum	A	7	NULL	NULL	YES
BTREE			YES	NULL					
orders	1	index_order	2	odate	A	7	NULL	NULL	YES
BTREE			YES	NULL					